

Week 1: block trades

Mathematical idea: The same boundary can have different decompositions. A color restriction links two constructions; area helps explain a minimum.

Use these as building mats. Allow free play before handing out the first task. Give one page at a time; the later pages are invitations, not a completion requirement. Children may point, speak, or build instead of writing.

Print and prepare

Print student packets **single-sided on US Letter at 100% / Actual Size**. Turn off “Fit” or “Shrink.” Each mat assumes a **1-inch (25.4 mm) block edge**. Compare the check line with an actual green triangle before the session. If your blocks differ, trace around the actual blocks on blank paper, or change `\blockside` in `common.tex` and rebuild; recheck page fit.

The K–1 packet has 2 pages; grades 2–3 and 4–5 have 3 each. For the current roster, print 3 K–1 packets, 3 middle packets, and 1 upper packet (18 student sheets). Keep this answer guide with the adults.

Per child, aim for **2 yellow, 4 red, 6 blue, and 8 green blocks**, with spare greens. Use only the green equilateral triangle, blue 60-degree rhombus, red trapezoid, and yellow hexagon. Set squares and thin rhombi aside. Add pencils, scrap paper, and optional sleeves. No color printing is needed: outlines are black and white. If colors are difficult to distinguish, use the shape names or G/B/R/Y labels; treat the restriction as “no block type in both copies.”

Choose an entry point

- K–1** Match edges and turn pieces; no reading, writing, or arithmetic required. Start with a single blue or red outline. Count only if useful.
- 2–3** Track block types across two copies; small counts are optional. Read prompts aloud as needed. Allow shared colors first, then fix one shared type at a time.
- 4–5** Add small piece counts and compare an example with a claim about every possible pair. No area formulas or multiplication tables required. Enter through the middle task if helpful.

A flexible hour

0–10: Free building. **10–15:** Trace or show the blue outline; remove the block and ask, “Can you change the pieces without changing the shape?” Check one complete trade together. **15–35:** Start each group’s first page. **35–40:** Move/reset. **40–55:** Continue a good investigation or offer the next page. **55–60:** Show one discovery and tidy up.

Keep one parent mainly with K–1. The organizer launches the middle group and the fifth grader, then returns to each regularly. Give each child a set of blocks and building space. Reserve a conversation with the fifth grader about whether a smaller pair is possible.

Play exchange shop: A partner brings a block; the shopkeeper trades it for smaller pieces covering the same space. Check the trade, then swap roles.

Useful questions: What have you tried? Did the outside change? Which colors must the other copy avoid? Can changing a group of pieces help? Let children demonstrate checks before supplying a hint.

Checked constructions

Diagram key: G = green triangle; B = blue rhombus; R = red trapezoid; Y = yellow hexagon. The diagrams below are reduced size. Student mats show only the outside boundary.

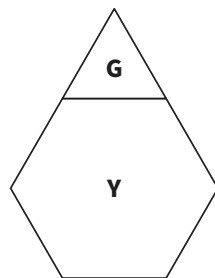
K-1 trades / F01-K

One blue rhombus trades for **two greens**. One red trapezoid trades for **one blue and one green**, or **three greens**. A fair trade covers exactly the same space. Place the original block over a replacement to compare. Turning an unchanged single piece does not make a new decomposition.

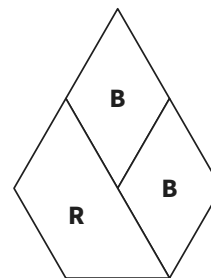
For the pennant, yellow + green and two reds + green give two fillings; shared colors are allowed at this level. Split the yellow hexagon into its top and bottom red trapezoids to make the second filling.

P1: the pennant / F01-M and F01-U

The outline is a hexagon with a green triangle attached outside one full edge. A legal pair uses **yellow + green / red + two blues**: $2 + 3 = 5$ pieces.



Copy A: Y + G

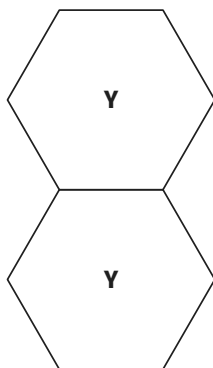


Copy B: R + B + B

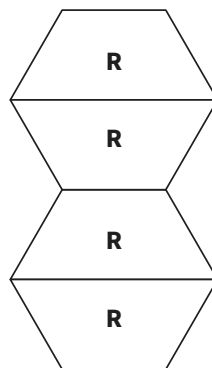
Hint ladder: Keep A as yellow + green. B may use only red and blue. Join the outside triangle to the inside triangle that touches its base: together they form a blue rhombus. The remaining region takes a red and a blue, as drawn.

P2: double hexagon / F01-M and F01-U

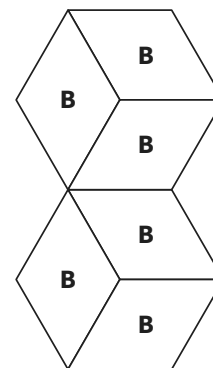
The two hexagons share one full edge. With yellow allowed, use **two yellows / four reds** (6 total). With yellow banned, use **four reds / six blues** (10 total). The same red construction works in both pairs.



2 yellows



4 reds



6 blues

Hint: Fill each hexagon separately, then keep the outside and change the pieces. Check both the geometric fit and the color rule.

Why fewer cannot work

Let the units come from the blocks

After children have tried, cover each block with green triangles. Their areas, measured in green triangles, are **G = 1**, **B = 2**, **R = 3**, **Y = 6**. P1 has area **7**; P2 has area **12**. Children can make the following arguments with blocks or drawings. A correct area sum alone does not prove a tiling exists; the constructions on page 2 supply that check.

P1: the minimum is 5 pieces across the pair

Each copy needs at least two pieces because even the largest block covers only 6 triangle units. To cover 7 with two blocks, the only possible areas are $6 + 1$: yellow + green. A four-piece pair would therefore need yellow + green in *both* copies, breaking the color rule. At least 5 pieces are needed, and the illustrated pair uses 5.

P2: the minimum is 6 pieces across the pair

If one copy uses yellow, it needs at least 2 pieces. The other copy cannot use yellow, so each of its pieces covers at most 3 triangle units. That copy needs at least 4 pieces. Total: at least $2 + 4 = 6$.

If neither copy uses yellow, each needs at least 4 pieces, already at least 8 total. Thus no pair can beat 6; two yellows / four reds achieves 6.

P2 without yellow: the minimum is 10 pieces

At most one copy may use red. If a copy uses red, its pieces cover at most 3 units each, so it needs at least 4. The other copy has only blue and green, covering at most 2 units per piece, so it needs at least 6. Total: at least 10. If neither copy uses red, each needs at least 6 pieces, giving at least 12 total. Four reds / six blues attains 10.

Extensions, recording, and sources

Keep an open question open. A child's new outline may have an interesting construction without a proven minimum. Record "best so far" until there is a reason that every smaller count fails. Children may create an outline, verify two fillings, and exchange just the traced boundary.

Prior use: Lowell Handout 3, problem 3.4, used hexagon decompositions; Handout 4, problems 4.3–4.5, and *Pattern Block Exercises*, problems 2–4 (PDF pp. 1–2), used trapezoid tilings. Give returning children the new P1/P2 constraints or minimization work. After the session, record the actual trades, P1/P2 pages, color restrictions, and arguments encountered by each child in `plans/fall-k-5-year-a-use-log.md`. Preparing these sheets does not mark them as taught.

Activity sources: JRMF, *Changing Colors Activity Guide*, PDF pp. 3–6, supplies simultaneous copies with no shared colors and an exploration-first launch. Its *Beginner Version*, PDF pp. 3–6, supplies different fillings without the color restriction. P1/P2 and the minimum-piece sequence come from this repository's *Fall Year A activity guide*, Week 1, F01-K/M/U; they are local constructions, not copied JRMF task sheets.

Format source: *Math Circle by the Bay*, preface, printed pp. ix–x (PDF pp. 10–11), describes manipulatives, extra challenges, practicing explanations, and adjusting depth across ages. Our one-page-at-a-time packets and two-adult hour are local adaptations, not a schedule prescribed by that book.